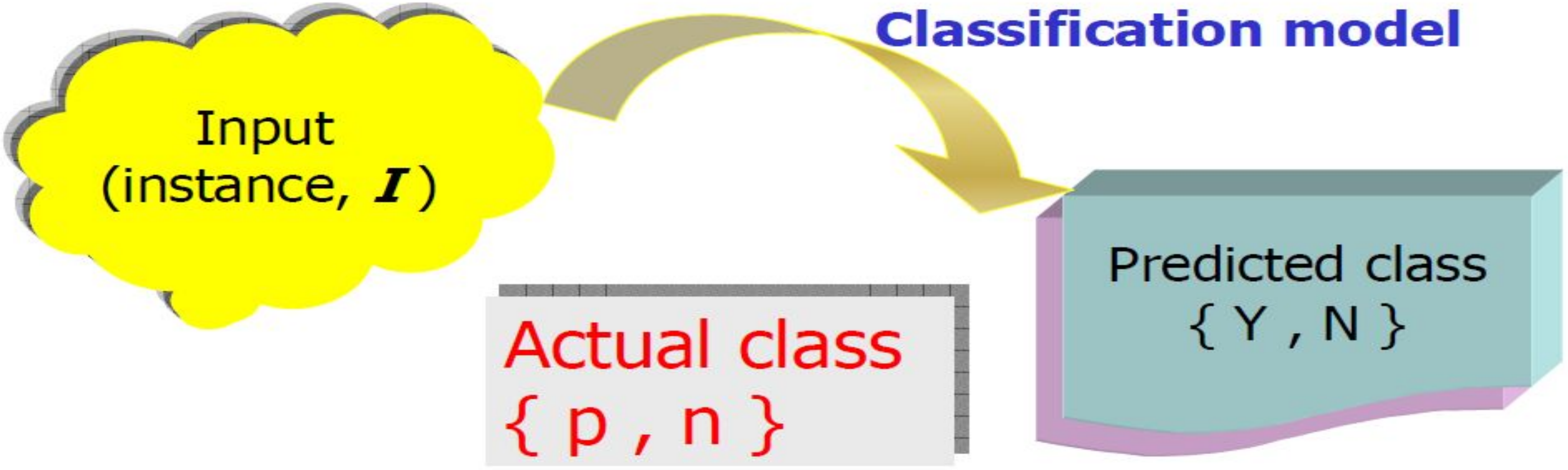
Performance Analysis and ROC Curve

Classier Performance

- Problem: two classes classification



PS: actual class {p: positive class, n: negative class}

Confusion Matrix

- Given a classifier and an instance:

Classifier Predicted class	TRUE CLASS	
	p (positive)	n (negative)
Y	True Positives	False Positives
N	False Negatives	True Negatives
Total	P	N

P = True Positives + False Negatives

Measuring Classifier Performance

→ If the instance is positive and it is classified as positive, it is counted as a **true positive**.
→ if the instance is positive and it is classified as negative, it is counted as a **false negative.**
→ If the instance is negative and it is classified as negative, it is counted as a **true negative.**
→ If the instance is negative and it is classified as positive, it is counted as a **false positive.**

Example- Confusion Matrix

Cancer Blood Test

True Positive

- Actually have it and test also says so

True Negative

- Don’t have it and test also says so

False Positive

- Have it but test doesn’t say so

False Negative

- Don’t have it but test says you have it.

Performance index

TP	FP
FN	TN

$$TPR = \frac{TP}{P} = Recall,\ FPR = \frac{FP}{N}$$

$$Precision = \frac{TP}{TP+FP},\ Accuracy = \frac{TP+TN}{P+N}$$

$$Sensitivity = Recall,\ Specificity = 1 - FPR$$

Performance Metrics

$$\text{fp rate} = \frac{FP}{N} \qquad \text{tp rate} = \frac{TP}{P}$$

$$\text{precision} = \frac{TP}{TP+FP} \qquad \text{recall} = \frac{TP}{P}$$

$$\text{accuracy} = \frac{TP+TN}{P+N} \qquad \text{F-measure} = \frac{2}{1/\text{precision}+1/\text{recall}}$$

$$\begin{aligned} \text{sensitivity} &= \text{recall} \\ \text{specificity} &= \frac{\text{True negatives}}{\text{False positives} + \text{True negatives}} \\ &= 1 - \textit{fp rate} \end{aligned}$$

→ **tp (true positive) rate / Recall / Hit rate / Sensitivity:** The proportion of positives that are correctly identified.
→ **fp (false positive) rate:** The proportions of negatives that are incorrectly identified.
→ **Precision / Positive predicted value**: The proportion of positively identified that are correct.
→ **F-measure:** A measure that combines precision and recall. It is the harmonic mean of precision and recall.
→ **Specificity:** The proportion of negatives that are correctly identified.

Receiver Operating Characteristic :

- is a graph showing the performance of a classification model at all classification thresholds.
- This curve plots two parameters:

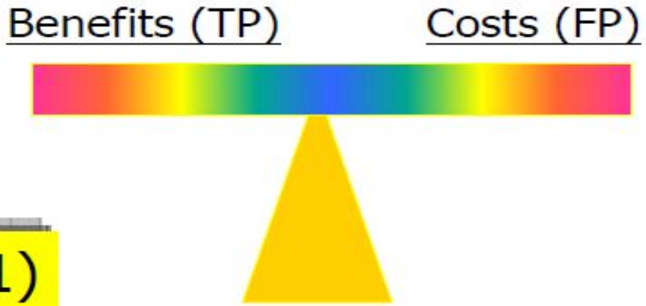
● True Positive Rate

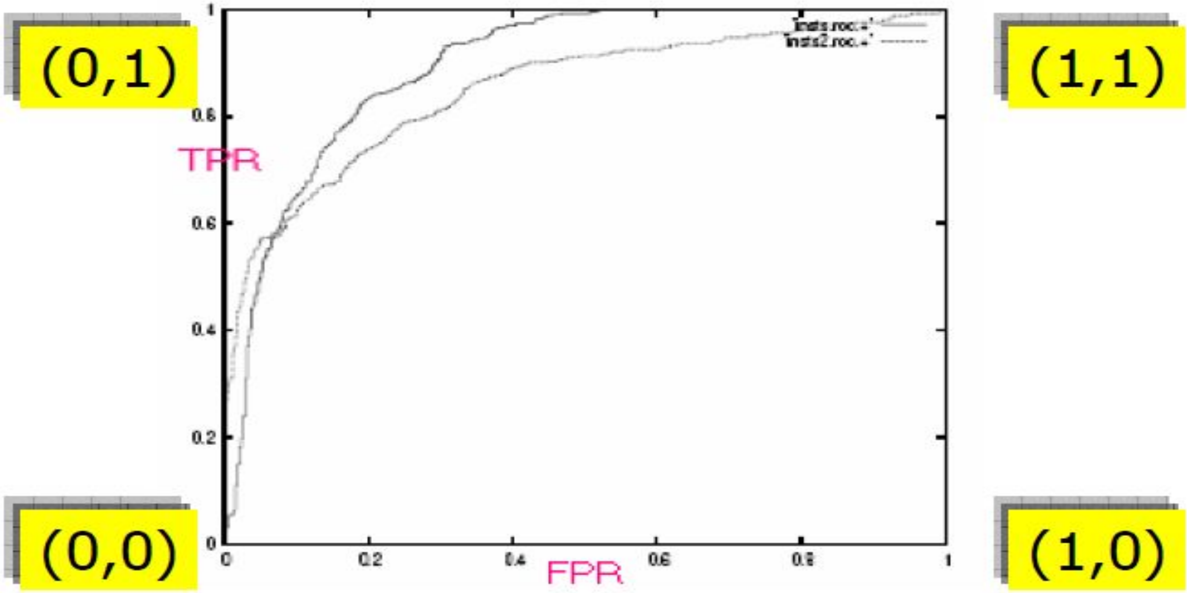
● False Positive Rate

- An ROC curve plots TPR vs. FPR at different classification thresholds.
- Lowering the classification threshold classifies more items as positive, thus increasing both False Positives and True Positives.

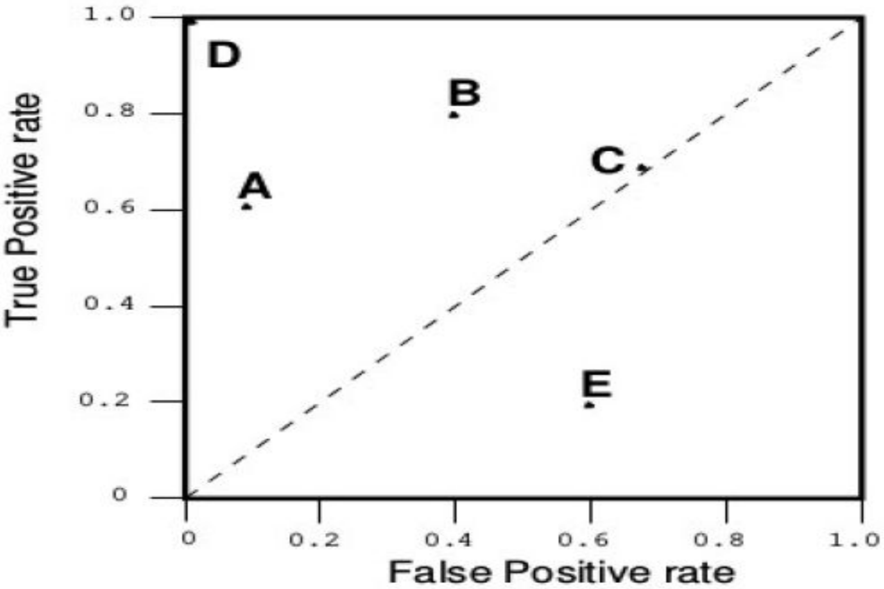
ROC curve

- Y axis: TPR
  X axis: FPR



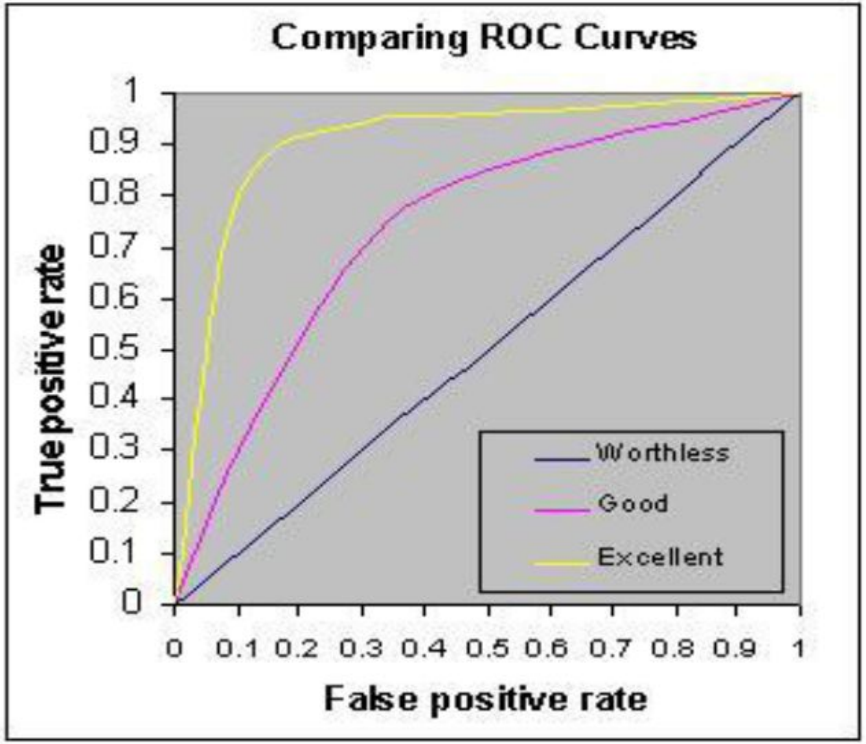


ROC Graph



→ One point in ROC space is better than another if it is to the northwest (tp rate is higher, fp rate is lower, or both) of the first. (D is better than all other points)
→ Classifiers in the lower left part are known as conservative.
→ Classifiers in the upper right part are known as liberals.
→ the diagonal y=x line represents a random classifier.
→ Classifiers in the lower right triangle performs worse than random classifier.

- .90-1 = excellent (A)
- .80-.90 = good (B)
- .70-.80 = fair (C)
- .60-.70 = poor (D)
- .50-.60 = fail (F)

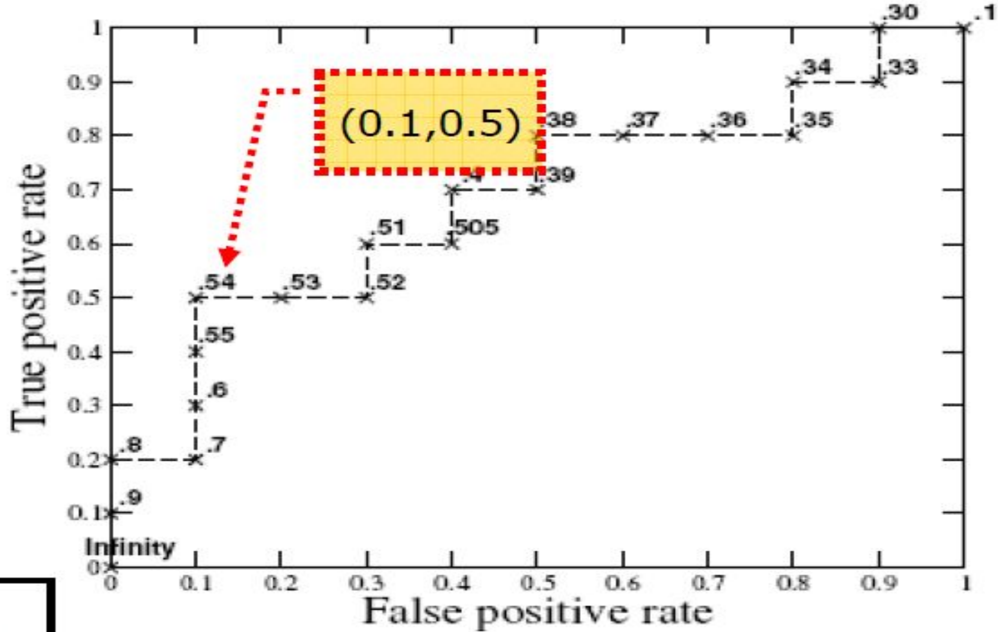


Use thresholds to create ROC curve

Inst#	Class	Score	Inst#	Class	Score
1	p	.9	11	p	.4
2	p	.8	12	n	.39
3	n	.7	13	p	.38
4	p	.6	14	n	.37
5	p	.55	15	n	.36
6	p	.54	16	n	.35
7	n	.53	17	p	.34
8	n	.52	18	n	.33
9	p	.51	19	p	.30
10	n	.505	20	n	.1

If threshold =0.54 →
Numbers of Score ≧0.54 → 6

5	1	6
5	9	14
10	10	20



$$x : \frac{1}{10} = 0.1$$

$$y : \frac{5}{10} = 0.5$$

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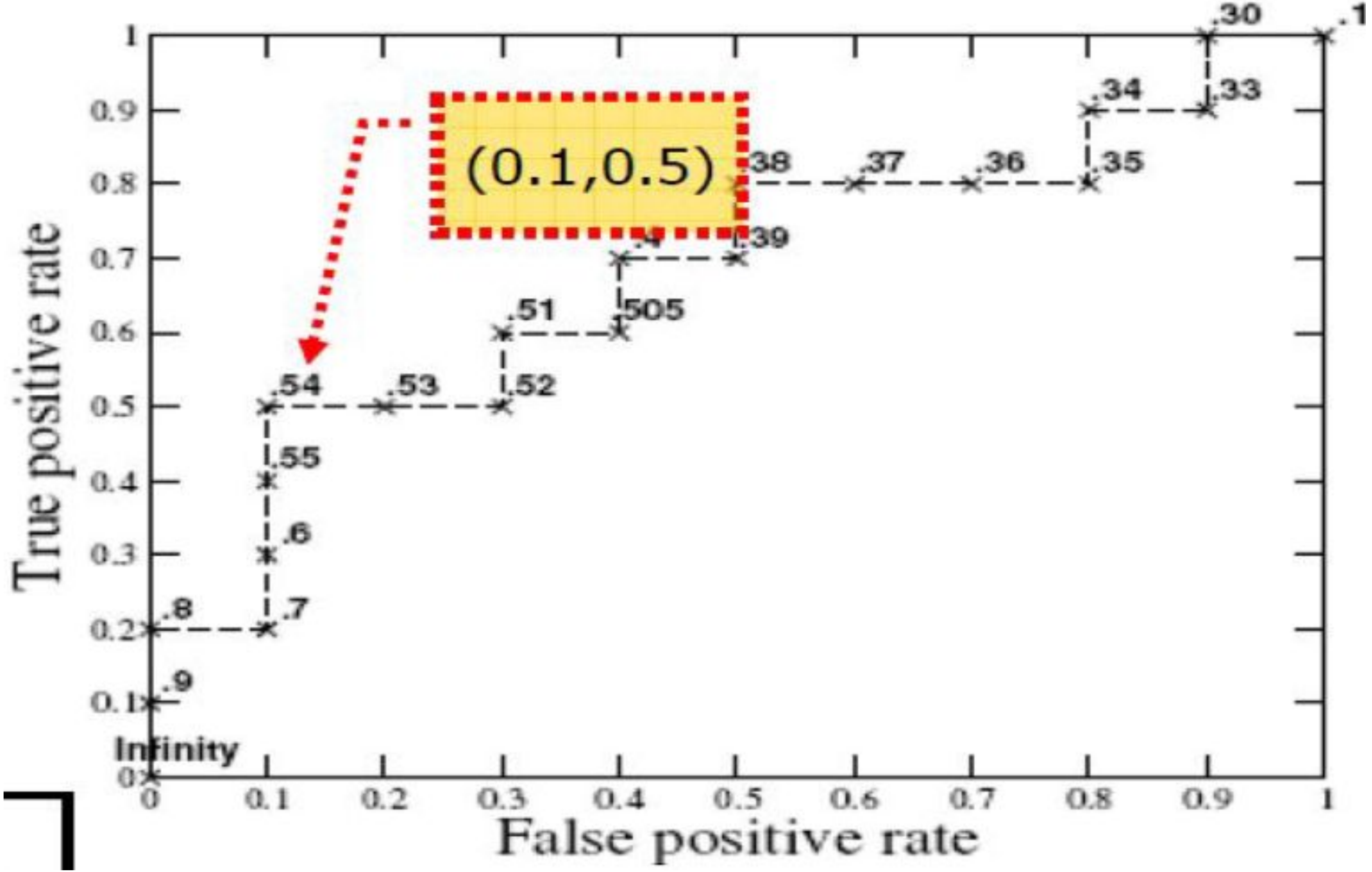
Threshold - 0.54

	p	n	
Threshold	5	1	6
Not Threshold	5	9	14
	10	10	20

FPR = 1/10 = 0.1

TPR = 5/10 = 0.5

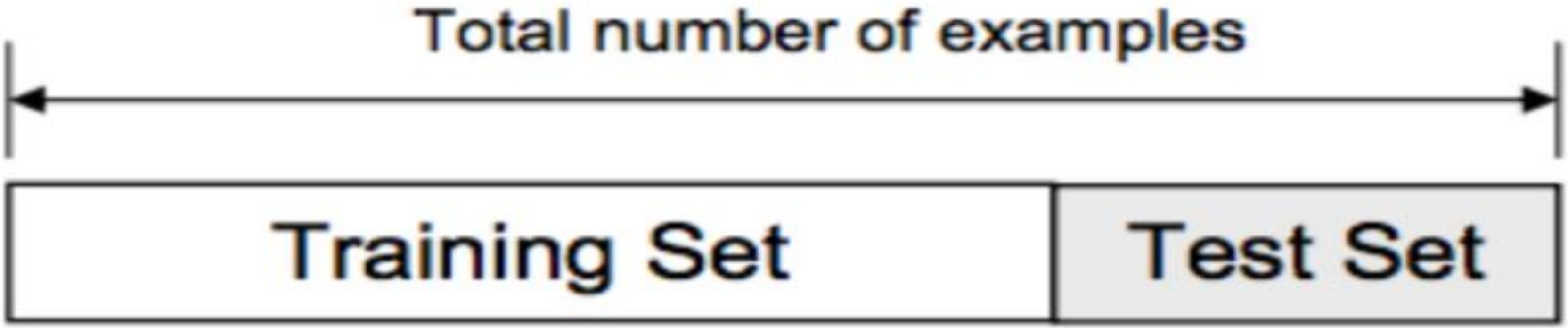


(0.1,0.5)
True positive rate
False positive rate
Infinity
.9
.8
.7
.6
.55
.54
.53
.52
.51
.505
.4
.39
.38
.37
.36
.35
.34
.33
.30
.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

Cross-validation

Cross-validation, it’s a model validation techniques for assessing how the results of a statistical analysis (model) will generalize to an independent data set.

- Helps to avoid overfitting,underfitting



K- Fold Cross-validation